Fiscal Year:	FY 2007	**Task Last Updated:**	FY 03/08/2008
PI Name:	Roberts, Donna M.D.		
Project Title:	Human Cerebral Cortex Plasticity in Response to Long-term Bedrest as an Analog to Microgravity		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	29425-8905	**Congressional District:**	6
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	03-OBPR-05
Start Date:	07/01/2004	**End Date:**	08/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Corrected end date to 8/30/2007; No-cost extension was until 8/30/2007 per PI (9/07) NOTE: Received no-cost extension to 06/30/2007 per PI (5/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ramsey, David (South Carolina Research Authority)		
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Task Description:	Understanding the factors governing human performance in microgravity is crucial to both mission planning and as a precursor to potential performance enhancement techniques. Preliminary data from our laboratory and others, suggest that alterations in cortical excitability occur in models that parallel microgravity. This research focuses on the general hypothesis that cortical motor map reorganization occurs in response to a microgravity environment and that these changes can be evaluated, and potentially modulated in an environment that models the physiological impact of microgravity (90 days extended bedrest). The first objective will be to measure cortical plasticity using cortical excitability as a metric and quantified by Transcranial Magnetic Stimulation (TMS) recruitment curves. This will be done by evaluating lower limb neurophysiological response to the application of TMS over the duration of the bedrest study. A second objective is to determine if repetitive TMS (rTMS) stimulation over motor cortex alters the rate or degree of plasticity when applied during a second bedrest campaign and measured identically to the first. A new set of TMS recruitment curves will be compared to the first, to determine if the stimulation modulated the time course of cortical excitability changes or impacts the length of the recovery period. Finally, we will correlate our measures of cortical excitability with NASA measures of lower limb functional recovery to quantify funtional outcomes in response to rTMS application.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Results of this research include a better understanding of the basic neurophysiology underlying cortical plasticity occurring during lower extremity disuse and suggest a possible intervention for remediating astronaut performance prior to landing on a planetary surface. On Earth, these data may lead to applications in modulating cortical maps furthering the rehabilitation of patients suffering from brain injury or prolonged confinement to bed. In fact, in our lab, we are applying this data to the development of a new methodology for the rehabilitation of lower extremity dysfunction. The technique may serve as adjunctive therapy to traditional methods of rehabilitation in stroke patients.
Task Progress:	Human performance in microgravity is crucial to the success of long-term space station and interplanetary missions. The NASA Bioastronautics Roadmap has identified several potential risks resulting in impaired astronaut performance. Impaired sensory-motor capability adversely affecting performance of operational tasks during and following missions throughout re-adaptation is a potential risk (Risks #13 and #14). Furthermore, human performance failure may occur due to conditions such as neurobehavioral and cognitive problems (Risk #25). Human performance failure may also occur due to a mismatch between crew cognitive capabilities and task demands (Risk #26). To assure maximum performance capability in the space environment, the acute and long-term effects of weightlessness on the central nervous system and specifically the sensory-motor system must be investigated. The purpose of this study was (1) to investigate motor cortex plasticity induced by a long-term bedrest model of microgravity, (2) to correlate cortical plasticity with alterations in performance using NASA standard measures of performance, and (3) to purposefully modulate cortical maps to ameliorate the known degradation in performance that occurs during periods of disuse. The end-product would be the development of a rehabilitative strategy for astronauts on long-term missions that would be suitable for use on-board the spacecraft to ensure maximal performance upon landing on a planetary surface. As an initial step in this direction, we used transcranial magnetic stimulation (TMS) to evaluate motor cortex plasticity in a long-term bedrest model. We hypothesized that alterations in afferent signaling, as a result of disuse, would induce cortical reorganization by altering cortical excitability. We additionally hypothesized that altered cortical excitability correlates with motor function and that cortical excitability can be modulated in useful ways to enhance performance. Our intention was also to favorably alter cortical plasticity in order to improve post-bedrest performance leading to the development of a countermeasure to microgravity-induced performance degradation. Our specific aims and the progress toward each aim are: Specific aim #1: To evaluate cortical plasticity in response to 60-90 days of bedrest. This experiment tested the following hypothesis: 1. Sixty to ninety days of strict bedrest will induce changes in the neurophysiological properties of lower extremity motor cortex as measured by TMS-derived recruitment curves. These changes will be maximal 24 hours following return to normal activity. Status: Completed. Specific aim #2: Using the results obtained during the first bedrest campaign, we would then test whether it is possible to alter bedrest induced cortical plasticity in a useful way to ameliorate the degradation in performance that occurs during periods of disuse. Status: Not accomplished, due to early termination of study, however based on the data that we have acquired, we could complete this aim if given access to 16 additional subjects. Specific aim #3: To collaborate with NASA investigators to correlate our measures of cortical excitability with their measures of musculoskeletal lower limb functionality. The following hypothesis was tested. 1. Measures of cortical excitability will correlate with measures of lower limb functional recovery. Status: Completed but need more subjects for statistical significance. As in our original grant proposal, we had intended to carry out specific aim #1 in the first bedrest campaign. According to our original power analysis, we hypothesized that a sample size of 18 would be required for a power of 0.90 and an alpha of 0.05 for a two-tailed t-test, effect size 1.85 to detect changes in cortical plasticity in the long-term bedrest model. We were asked to participate in bedrest campaigns 3a and 3b. In campaign 3a, we were provided with 4 subjects that completed the protocol. In campaign 3b, we were provided with an additional 5 subjects, however, due to a nature disaster (Hurricane Rita in the wake of Hurricane Katrina in the fall of 2005) this campaign was abruptly terminated early without an opportunity to acquire the necessary post-bedrest data. Therefore, we were provided with a total of four subjects that carried out our experimental protocol towards specific aims #1 and #3 and the results we are reporting here are based on the data from these four subjects.

	Data from four subjects is not sufficient to complete this study. We could carry out specific aim #2, leading to countermeasure development, but would require access to an additional 16 subjects.
Bibliography Type:	Description: (Last Updated: 01/15/2026)
Abstracts for Journals and Proceedings	Johnson KA. "Mapping Cortical Plasticity in the Bedrest Analogue of Space Flight Using TMS and fMRI." Presented at the 12th Organization for Human Brain Mapping Annual Meeting, Florence, Italy, June 2006. Proceedings, Organization for Human Brain Mapping 12th Annual Meeting, Florence, Italy, June 2006. , Jun-2006
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Articles in Peer-reviewed Journals	Roberts DR, Vincent DJ, Speer AM, Bohning DE, Cure J, Young J, George MS. "Multi-modality mapping of motor cortex: comparing echoplanar BOLD fMRI and transcranial magnetic stimulation. Short communication." J Neural Transm. 1997;104(8-9):833-43. PMID: 9451716 , Mar-2003
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